

Blockchain-Enhanced Process Path Tracking using CDC and Kafka

Presented by:

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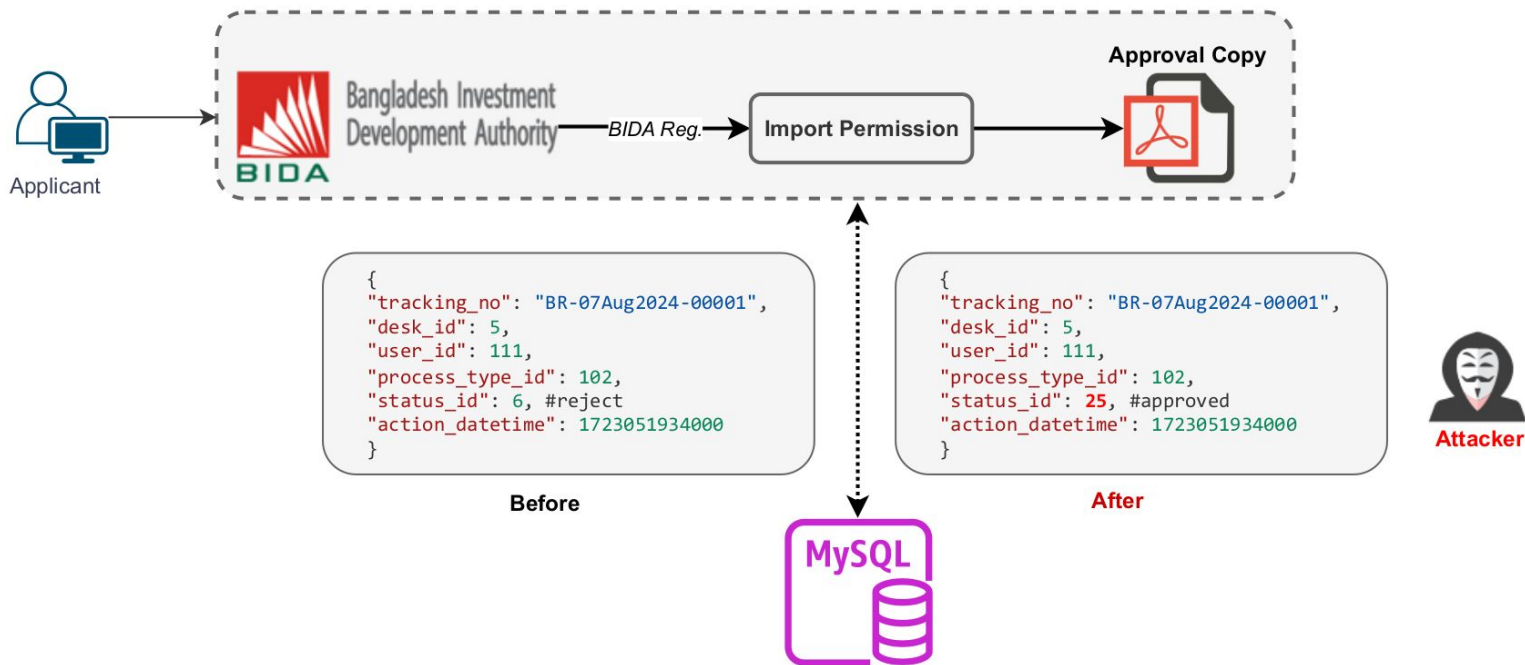
Machine Learning Engineer

WebCrafter Team

Project Background

- We aim to create a system that integrates blockchain with real-time data processing using Change Data Capture (CDC), Kafka, and Debezium.
- The goal is to ensure data integrity, transparency, and traceability across distributed systems. (detecting any manipulation among the process data)

The Problem



Technology Stack

Change Data Capture (CDC):

- Captures database changes (inserts, updates, deletes) in real-time.
- Methods include log-based, trigger-based, and timestamp-based CDC.

Apache Kafka:

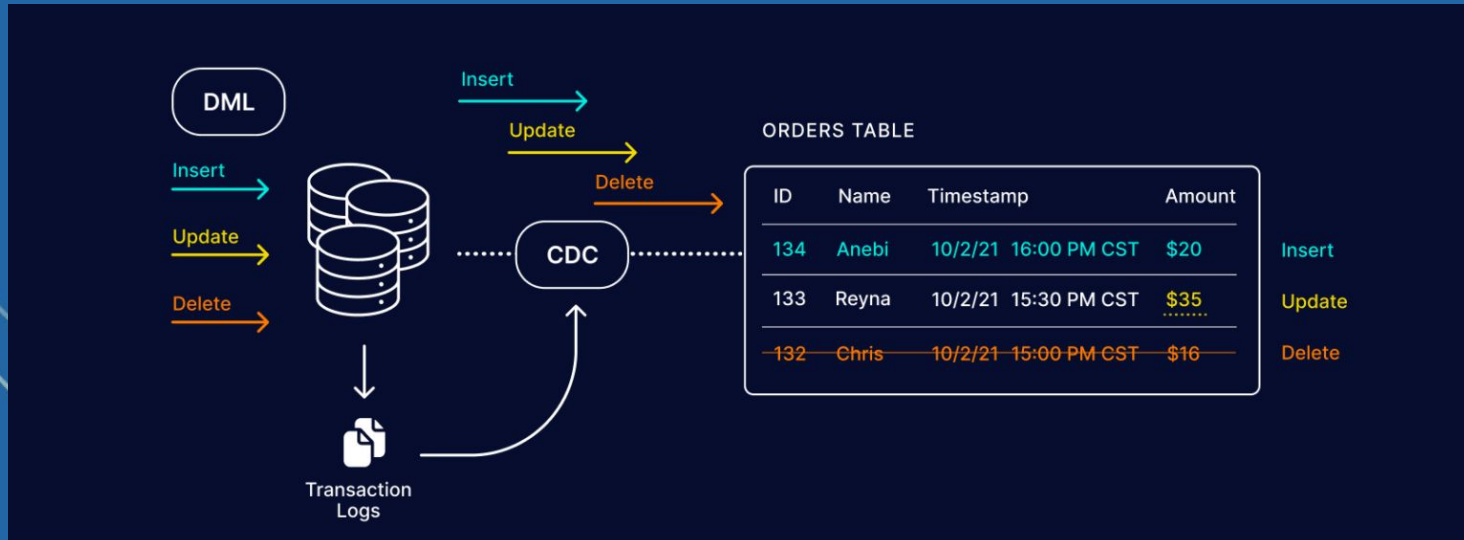
- A distributed messaging system that streams data from multiple sources.
- Handles large volumes of data with high fault tolerance.

Debezium:

- A CDC tool that monitors databases and captures changes.
- Integrates with Kafka to publish changes to specified topics.

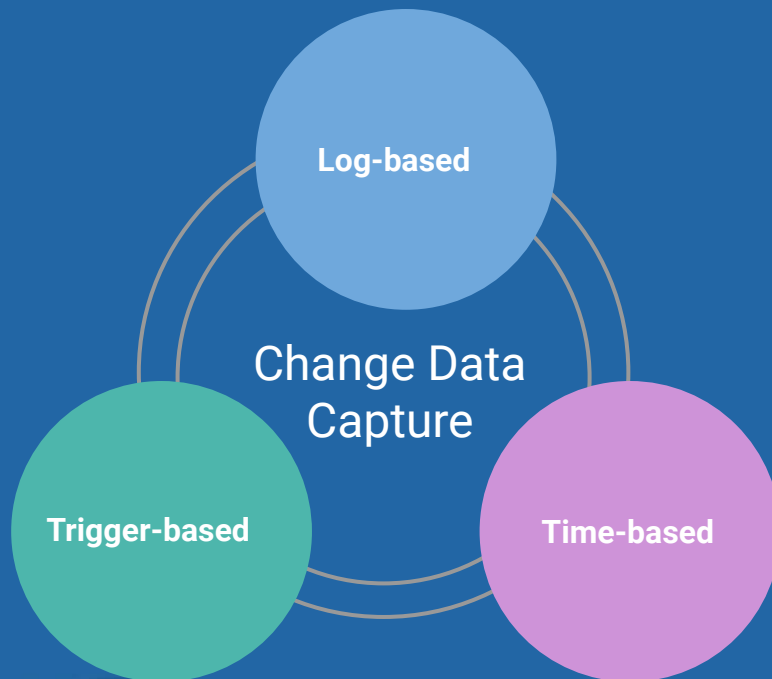
Change Data Capture (CDC)

Change data capture (CDC) refers to the process of identifying and capturing changes made to data in a database and then delivering those changes in real-time to a downstream process or system.



Change Data Capture (CDC)

There are three types of methods for implementing CDC.



Change Data Capture (CDC)

Log-Based CDC:

- Databases contain transaction logs (also called redo logs) that store all database events allowing for the database to be recovered in the event of a crash

Trigger-Based CDC

- Databases have trigger functions, which are automated stored procedures activated upon specific events (e.g., `INSERT`, `UPDATE`, or `DELETE`)

Time-Based CDC

- Some applications record when every row change is performed in a separate column (e.g., `LAST_MODIFIED`)

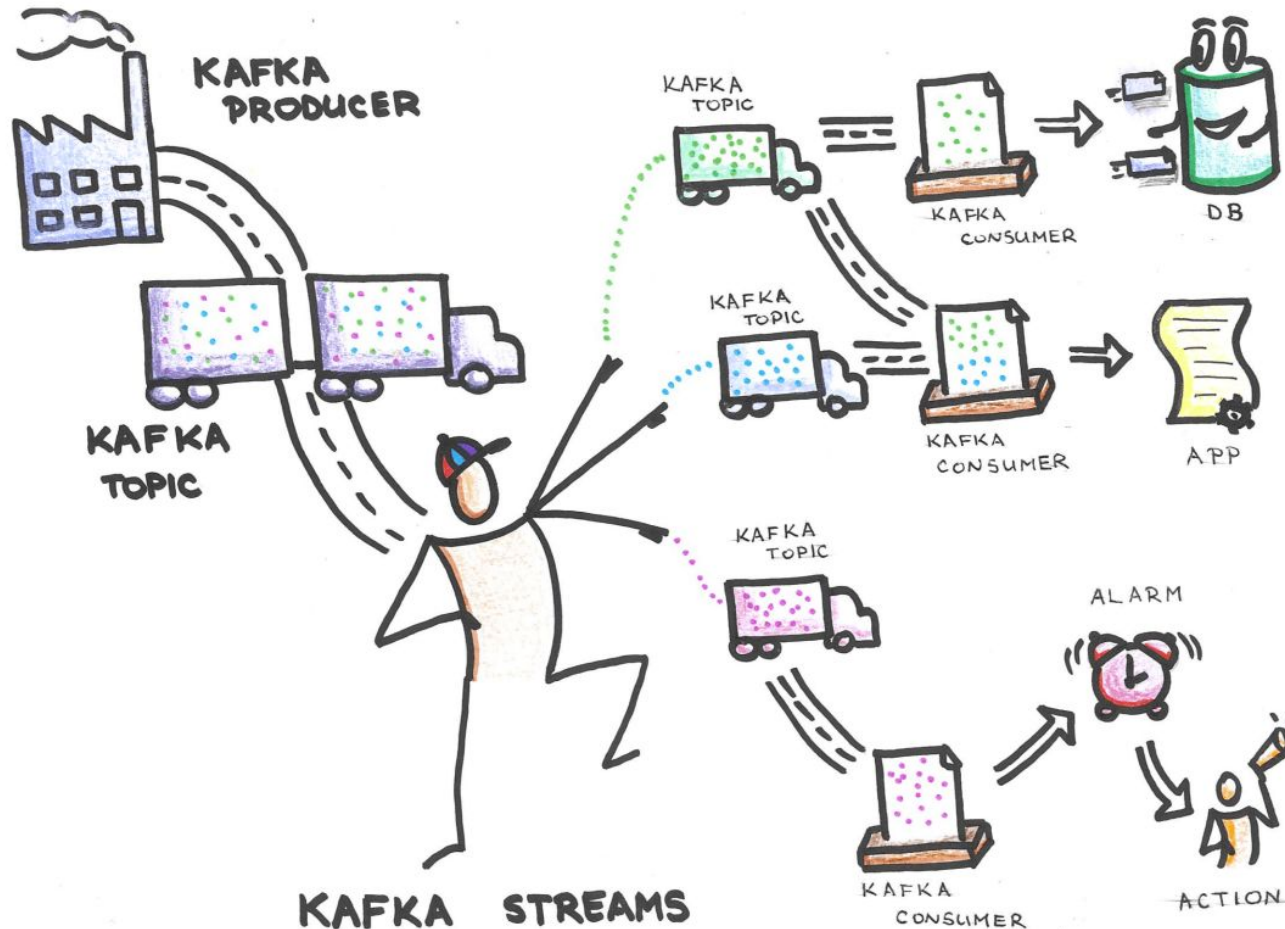
Apache Kafka

Apache Kafka is a distributed event streaming platform that is used for building real-time data pipelines and streaming applications

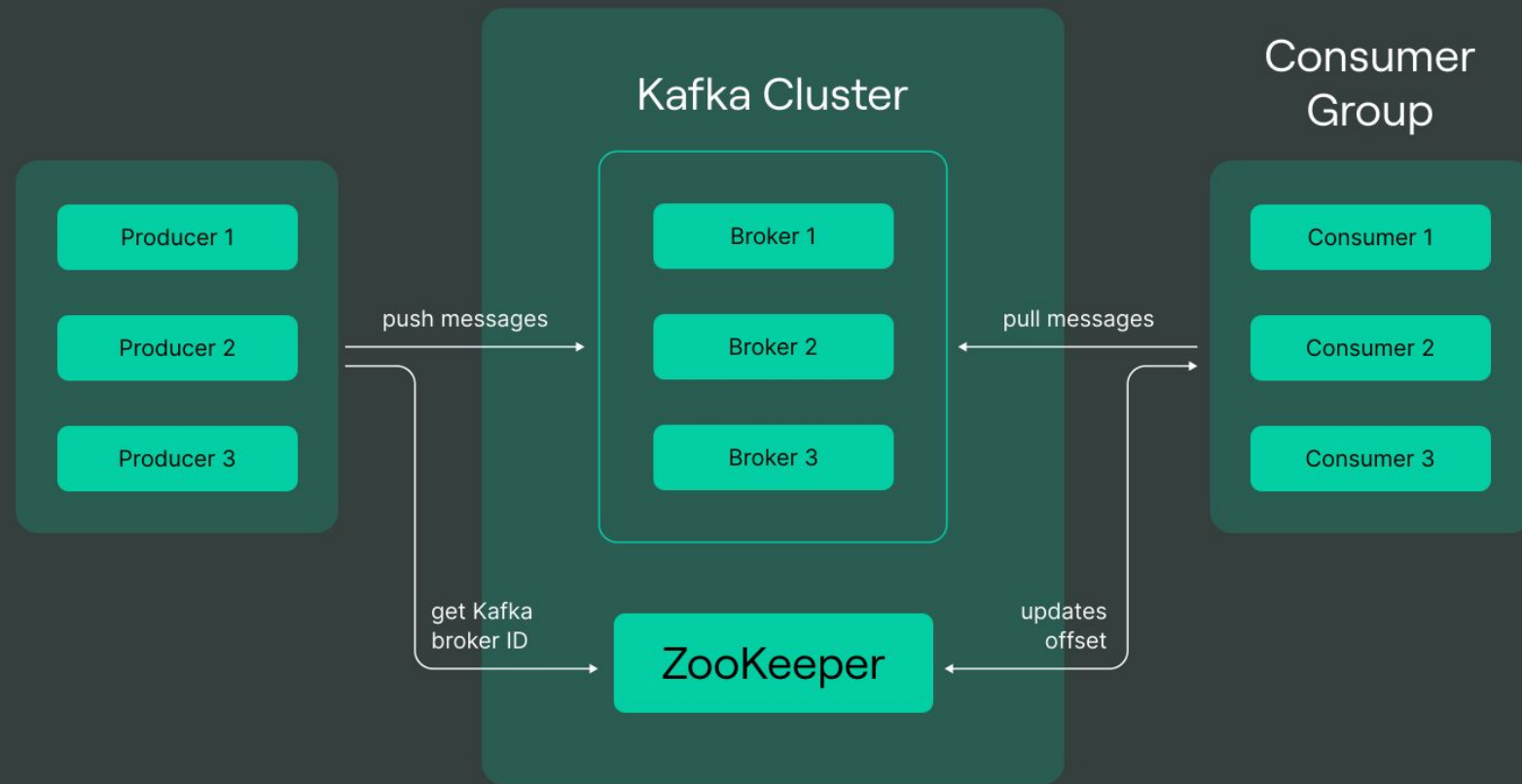
Kafka is a distributed publish-subscribe messaging system. Data is written to Kafka topics by producers and consumed from those topics by consumers

Apache Kafka

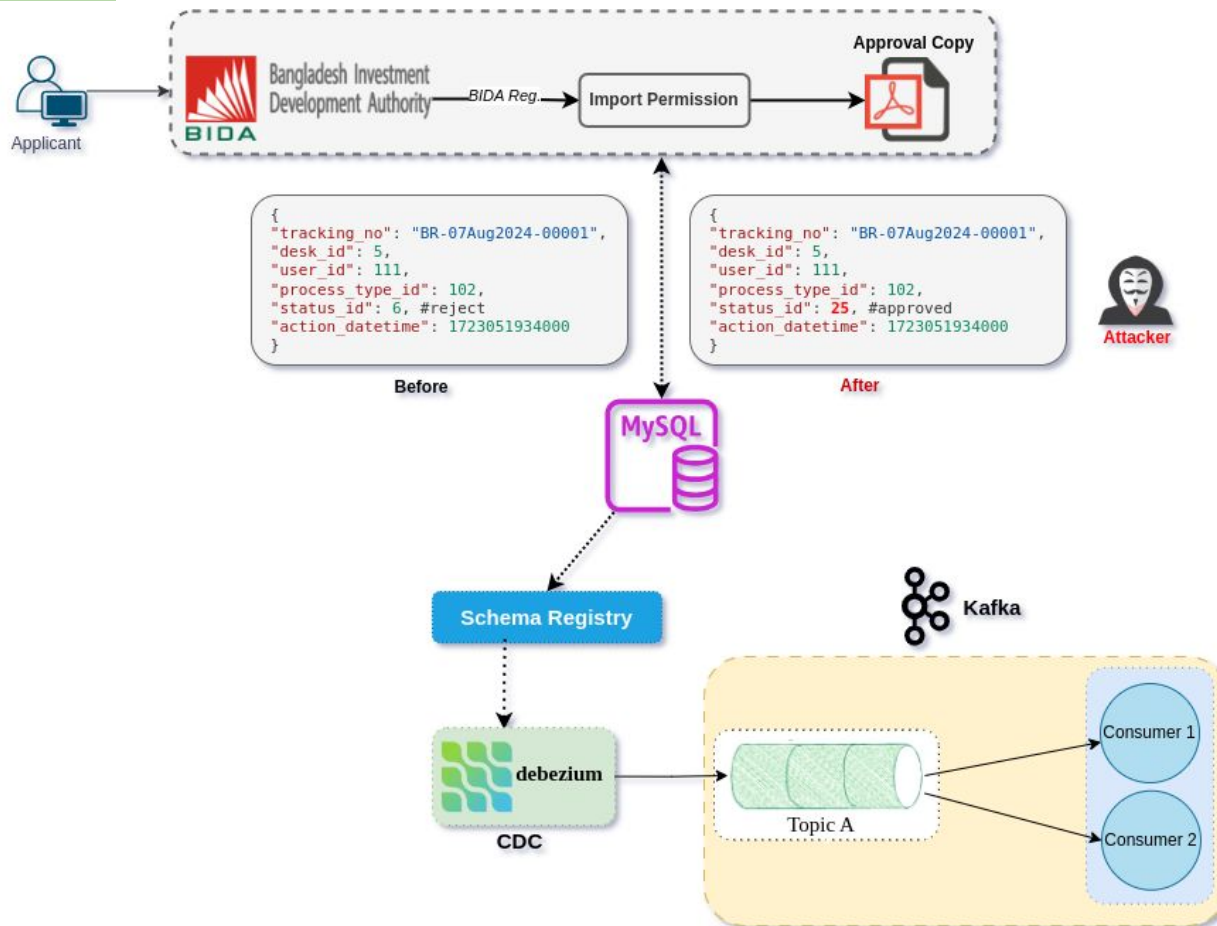
business
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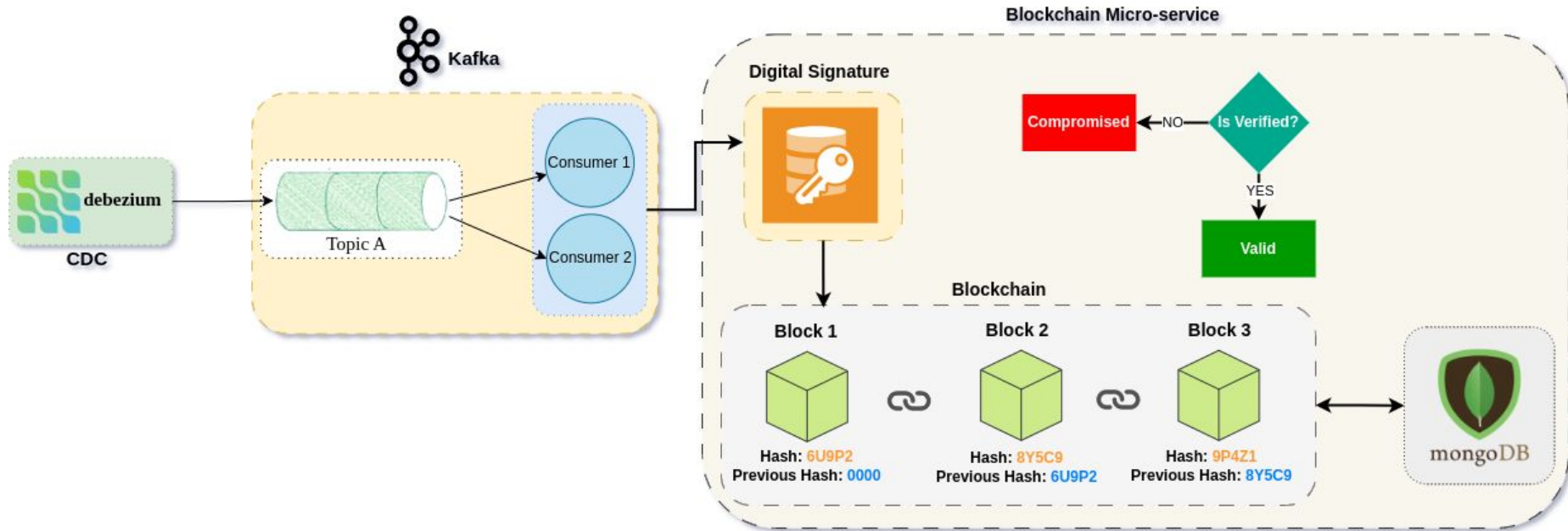
Kafka Ecosystem



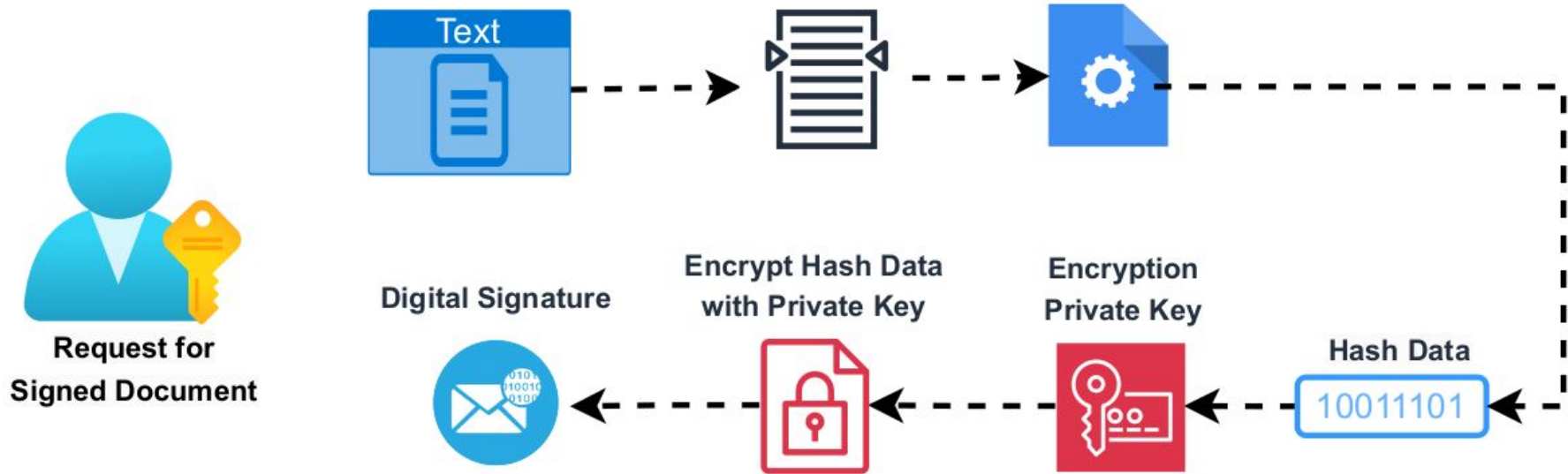
Our Solution



Our Solution



Digital Signature Service



Step 1: Data Generation from BIDA Services

- Data is generated from various BIDA services (e.g., form submissions, transactions).

Step 2: Change Detection (CDC)

- CDC captures changes in the source databases within BIDA.
- Changes are structured into events for further processing.

Step 3: Kafka Ingestion

- CDC events are published to Kafka topics.
- Kafka acts as the central data pipeline, streaming events in real-time.

Step 4: Digital Signature Services

- Kafka events are sent to Digital Signature Services.
- Digital signatures are applied to ensure data authenticity and integrity.

Step 5: Blockchain Storage

- Signed events are recorded onto the blockchain.
- Blockchain ensures immutability and traceability of the data.

Step 6: Interaction with MongoDB

- The blockchain interacts with MongoDB for data retrieval and additional processing.
- MongoDB serves as the database for storing and querying data.

Step 7: Verification and Monitoring

- Each block is verified and added to the chain.
- Continuous monitoring of the data stream, digital signatures, and blockchain status.

Use Cases and Applications

Financial Transactions:

- Real-time monitoring of transactions, ensuring compliance.

Regulatory Compliance and Auditability:

- Automated logging and reporting of transactions, ensuring regulatory compliance.

Voting Systems:

- Secure, transparent, and auditable voting processes with real-time results.

THANK YOU